

Calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

You have 85 minutes.

Signature

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

Solve the equation for solutions over the interval $[0, 2\pi)$. Show all work for credit (do not use solve). Give solutions to the thousandth. (3 points)

1. $4\sin^2 x + 2\sin x = 1$

2. Find the angle between $\mathbf{u} = \langle -3, 7 \rangle$ and $\mathbf{v} = \langle 5, 2 \rangle$ to the nearest thousandth degree.

3 pts

Calculator

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force and the angle the resultant makes with the 425-newton force. (4 points)

3. forces of 425 and 310 newtons, forming an angle of 68°

4. Find the force required to keep a 2400-lb car parked on a hill that makes an angle of 12° with the horizontal.
4 pts

Calculator

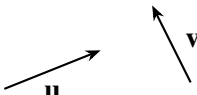
SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

5. A plane is headed on a bearing of 152° at an airspeed of 265 mph. A 35 mph wind is blowing from a direction of 215° . Find the groundspeed and resulting bearing of the plane. 4 pts

Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

6. Given vectors $\mathbf{u}$ and $\mathbf{v}$, sketch and clearly label a single geometric diagram representing: a) $\mathbf{u} + \mathbf{v}$ b) $2\mathbf{u} - \mathbf{v}$ 	2 pts each
7. Given vectors $\mathbf{u}$ and $\mathbf{v}$, find $\mathbf{u} = 3\mathbf{i} - 4\mathbf{j}, \quad \mathbf{v} = -2\mathbf{i} + 5\mathbf{j}$ a) $-3\mathbf{v}$ b) $2\mathbf{u} + \mathbf{v}$ c) $\mathbf{v} - \mathbf{u}$	5 pts
8. Find the magnitude and the positive direction angle for $\mathbf{u}$. $\mathbf{u} = \langle -2\sqrt{3}, 2 \rangle$	3 pts

9. For $\mathbf{u} = \langle -5, 2 \rangle$ and $\mathbf{v} = \langle 3, 8 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$.

2 pts

10. Determine whether the vectors are orthogonal.

2 pts

$$\mathbf{v} = 3\mathbf{i} + \sqrt{2}\mathbf{j}, \quad \mathbf{w} = \sqrt{2}\mathbf{i} + 3\mathbf{j}$$

Solve each equation algebraically on the given interval $[0, 2\pi)$. (3 points each)

11. $2 \sin 2x = -\sqrt{2}$

12. $\sqrt{3} \sin x = 2 \sin x \cos x$

Solve the equation algebraically on the given interval $[0^\circ, 360^\circ)$. (3 points)

13. $-\sec^2 \theta + 2 = \sec \theta$

Solve the equation algebraically on the given interval $[0^\circ, 360^\circ)$. (4 points)

14. $2\cos^2 \theta + \sin \theta = 1$

Find the exact value of each expression. (1 point each for 15 & 18 and 2 pts each for 16 & 17)

15. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$	16. $\cos^{-1}\left(\cos \frac{7\pi}{6}\right)$
17. $\sec^{-1}\left(\csc \frac{\pi}{6}\right)$	18. $\cot^{-1}(-\sqrt{3})$

Give the exact value of each expression. Rationalize the denominator of your answer. (3 points each)

19. $\csc\left(2\sin^{-1}\frac{1}{4}\right)$	20. $\sin\left(\tan^{-1}2 + \cos^{-1}\frac{3}{5}\right)$
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ANSWER KEY – Calculator**SHOW ALL WORK FOR FULL CREDIT!**

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandths.

Solve the equation for solutions over the interval $[0, 2\pi)$. Show all work for credit (do not use solve). Give solutions to the thousandth. (3 points)

1. $4\sin^2 x + 2\sin x = 1$

Set equal to 0; quadratic in $\sin x$ that does not factor, so use the quadratic formula.

$$4\sin^2 x + 2\sin x - 1 = 0 \implies \sin x = \frac{-2 \pm \sqrt{2^2 - 4(4)(-1)}}{2(4)} = \frac{-2 \pm \sqrt{20}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

$$\sin x = \frac{-1 + \sqrt{5}}{4} \approx 0.3090 > 0: \text{ sine positive in QI and QII. } x = \sin^{-1}(0.3090) \approx 0.314, x = \pi - 0.314 \approx 2.827.$$

$$\sin x = \frac{-1 - \sqrt{5}}{4} \approx -0.8090 < 0: \text{ sine negative in QIII and QIV. Reference angle } \sin^{-1}(0.8090) \approx 0.942.$$

$$x = \pi + 0.942 \approx 4.084, x = 2\pi - 0.942 \approx 5.341.$$

$$x \approx 0.314, 2.827, 4.084, 5.341$$

2. Find the angle between $\mathbf{u} = \langle -3, 7 \rangle$ and $\mathbf{v} = \langle 5, 2 \rangle$ to the nearest thousandth degree.

3 pts

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|} \quad \mathbf{u} \cdot \mathbf{v} = (-3)(5) + (7)(2) = -15 + 14 = -1$$

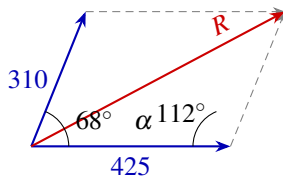
$$|\mathbf{u}| = \sqrt{(-3)^2 + 7^2} = \sqrt{58} \quad |\mathbf{v}| = \sqrt{5^2 + 2^2} = \sqrt{29}$$

$$\cos \theta = \frac{-1}{\sqrt{58}\sqrt{29}} = \frac{-1}{\sqrt{1682}} \approx -0.02438 \implies \theta = \cos^{-1}\left(\frac{-1}{\sqrt{1682}}\right) \approx 91.397^\circ$$

Negative dot product, so the angle had to come out obtuse (just past 90°). Calculator in degree mode.

Two forces act at a point in the plane. The angle between the two forces is given. Find the magnitude of the resultant force and the angle the resultant makes with the 425-newton force. (4 points)

3. forces of 425 and 310 newtons, forming an angle of 68°



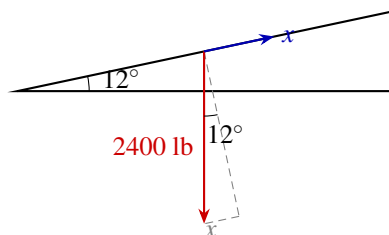
Complete the parallelogram. The angle across from the resultant is $180^\circ - 68^\circ = 112^\circ$ (consecutive angles of a parallelogram are supplementary). Law of Cosines for $|R|$, then Law of Sines for the angle α between R and the 425 N force (the 310 N side is across from α).

$$|R| = \sqrt{425^2 + 310^2 - 2(425)(310)\cos 112^\circ} = \sqrt{375433.8} \approx 612.727 \text{ N}$$

$$\frac{\sin \alpha}{310} = \frac{\sin 112^\circ}{612.727} \implies \sin \alpha \approx 0.46910 \implies \alpha \approx 27.976^\circ$$

The resultant has a magnitude of about 612.727 newtons and makes an angle of about 27.976° with the 425-newton force. (Do not assume the resultant bisects the 68° ; it leans toward the bigger force, so $\alpha < 34^\circ$ checks.)

4. Find the force required to keep a 2400-lb car parked on a hill that makes an angle of 12° with the horizontal. 4 pts

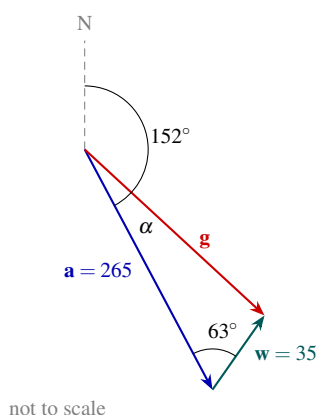


The weight (2400 lb, straight down) is the hypotenuse of the force triangle. The angle between the weight and the perpendicular to the hill equals the hill angle, 12° , so the component of the weight parallel to the hill sits *across* from 12° . That component is what the holding force x must match. SOH:

$$\sin 12^\circ = \frac{x}{2400} \implies x = 2400 \sin 12^\circ \approx \boxed{498.988 \text{ lb}}$$

A force of about 498.988 pounds is required to keep the car parked on the hill.

5. A plane is headed on a bearing of 152° at an airspeed of 265 mph. A 35 mph wind is blowing from a direction of 215° . Find the groundspeed and resulting bearing of the plane. 4 pts



$\mathbf{a} + \mathbf{w} = \mathbf{g}$ (airspeed plus wind equals groundspeed). Wind *from* 215° blows *toward* $215^\circ - 180^\circ = 35^\circ$. The headings of $\mathbf{a}$ and $\mathbf{w}$ differ by $152^\circ - 35^\circ = 117^\circ$, so the interior angle where $\mathbf{w}$ hangs off the tip of $\mathbf{a}$ is $180^\circ - 117^\circ = 63^\circ$. Law of Cosines for $|\mathbf{g}|$, Law of Sines for the drift angle α (across from $\mathbf{w}$).

$$|\mathbf{g}| = \sqrt{265^2 + 35^2 - 2(265)(35)\cos 63^\circ} = \sqrt{63028.5} \approx \boxed{251.055 \text{ mph}}$$

$$\frac{\sin \alpha}{35} = \frac{\sin 63^\circ}{251.055} \implies \sin \alpha \approx 0.12422 \implies \alpha \approx 7.136^\circ$$

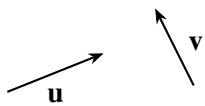
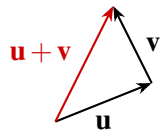
The wind blows toward the northeast, which swings the plane counterclockwise off its heading (toward a smaller bearing), so subtract:

$$\text{bearing} = 152^\circ - 7.136^\circ \approx \boxed{144.864^\circ}$$

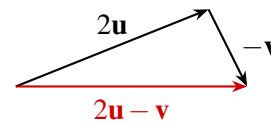
The groundspeed is about 251.055 mph and the resulting bearing is about 144.864° . (Groundspeed came out *less* than airspeed: the 117° spread means the wind has a headwind component.)

ANSWER KEY – Non-Calculator**SHOW ALL WORK FOR FULL CREDIT!**6. Given vectors $\mathbf{u}$ and $\mathbf{v}$, sketch and clearly label a single geometric diagram representing:

2 pts each

a) $\mathbf{u} + \mathbf{v}$ 

Tip to tail: slide $\mathbf{v}$ so its tail sits on the tip of $\mathbf{u}$; the sum runs from the tail of $\mathbf{u}$ to the tip of the moved $\mathbf{v}$.

b) $2\mathbf{u} - \mathbf{v}$ 

Double the length of $\mathbf{u}$ (same direction), then add $-\mathbf{v}$: same length as $\mathbf{v}$, opposite direction, tail on the tip of $2\mathbf{u}$.

7. Given vectors $\mathbf{u}$ and $\mathbf{v}$, find

5 pts

$$\mathbf{u} = 3\mathbf{i} - 4\mathbf{j}, \quad \mathbf{v} = -2\mathbf{i} + 5\mathbf{j}$$

a) $-3\mathbf{v}$ b) $2\mathbf{u} + \mathbf{v}$ c) $\mathbf{v} - \mathbf{u}$ Given in $\mathbf{i}, \mathbf{j}$ form, so answer in $\mathbf{i}, \mathbf{j}$ form.

$$-3(-2\mathbf{i} + 5\mathbf{j}) = \boxed{6\mathbf{i} - 15\mathbf{j}}$$

$$2(3\mathbf{i} - 4\mathbf{j}) + (-2\mathbf{i} + 5\mathbf{j})$$

$$= 6\mathbf{i} - 8\mathbf{j} - 2\mathbf{i} + 5\mathbf{j} = \boxed{4\mathbf{i} - 3\mathbf{j}}$$

$$(-2\mathbf{i} + 5\mathbf{j}) - (3\mathbf{i} - 4\mathbf{j})$$

$$= (-2 - 3)\mathbf{i} + (5 + 4)\mathbf{j} = \boxed{-5\mathbf{i} + 9\mathbf{j}}$$

8. Find the magnitude and the positive direction angle for $\mathbf{u}$.

3 pts

$$\mathbf{u} = \langle -2\sqrt{3}, 2 \rangle$$

$$|\mathbf{u}| = \sqrt{(-2\sqrt{3})^2 + 2^2} = \sqrt{12 + 4} = \sqrt{16} = \boxed{4}$$

$$\tan \theta = \frac{b}{a} = \frac{2}{-2\sqrt{3}} = -\frac{1}{\sqrt{3}} \Rightarrow \text{reference angle } 30^\circ$$

$$a < 0, b > 0 \text{ puts } \mathbf{u} \text{ in QII (tangent negative there checks), so } \theta = 180^\circ - 30^\circ = \boxed{150^\circ} \text{ or } \boxed{\theta = \frac{5\pi}{6}}.$$

The direction angle comes from the quadrant of the vector, never straight off $\tan^{-1}$.

9. For $\mathbf{u} = \langle -5, 2 \rangle$ and $\mathbf{v} = \langle 3, 8 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$.

2 pts

$$\mathbf{u} \cdot \mathbf{v} = (-5)(3) + (2)(8) = -15 + 16 = \boxed{1}$$

The dot product is a number, not a vector. Positive, so the angle between them is acute.

10. Determine whether the vectors are orthogonal.

2 pts

$$\mathbf{v} = 3\mathbf{i} + \sqrt{2}\mathbf{j}, \quad \mathbf{w} = \sqrt{2}\mathbf{i} + 3\mathbf{j}$$

Orthogonal means the dot product is 0. $\mathbf{v} = \langle 3, \sqrt{2} \rangle, \mathbf{w} = \langle \sqrt{2}, 3 \rangle$.

$$\mathbf{v} \cdot \mathbf{w} = (3)(\sqrt{2}) + (\sqrt{2})(3) = 3\sqrt{2} + 3\sqrt{2} = 6\sqrt{2} \neq 0$$

No, not orthogonal

Solve each equation algebraically on the given interval $[0, 2\pi)$. (3 points each)

11. $2 \sin 2x = -\sqrt{2}$

Isolate the trig function, then solve for the whole argument $2x$ first.

$$\sin 2x = -\frac{\sqrt{2}}{2}$$

Sine negative: QIII, QIV; reference angle $\frac{\pi}{4}$. General solution on $2x$:

$$2x = \frac{5\pi}{4} + 2\pi k \quad \text{or} \quad 2x = \frac{7\pi}{4} + 2\pi k$$

Divide everything by 2, the k term included:

$$x = \frac{5\pi}{8} + \pi k \quad \text{or} \quad x = \frac{7\pi}{8} + \pi k$$

$$k = 0: \frac{5\pi}{8}, \frac{7\pi}{8} \quad k = 1: \frac{13\pi}{8}, \frac{15\pi}{8} \quad k = 2: \text{too big.}$$

$$x = \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$$

12. $\sqrt{3} \sin x = 2 \sin x \cos x$

Never divide both sides by $\sin x$; that throws away solutions. Move everything to one side and factor the GCF.

$$\sqrt{3} \sin x - 2 \sin x \cos x = 0$$

$$\sin x (\sqrt{3} - 2 \cos x) = 0$$

$$\sin x = 0 \implies x = 0, \pi$$

$$\cos x = \frac{\sqrt{3}}{2} \implies x = \frac{\pi}{6}, \frac{11\pi}{6} \text{ (QI, QIV)}$$

$$x = 0, \frac{\pi}{6}, \pi, \frac{11\pi}{6}$$

Solve the equation algebraically on the given interval $[0^\circ, 360^\circ)$. (3 points)

13. $-\sec^2 \theta + 2 = \sec \theta$

One trig function, quadratic in form. Set equal to 0 with a positive leading term, then factor.

$$\sec^2 \theta + \sec \theta - 2 = 0 \implies (\sec \theta + 2)(\sec \theta - 1) = 0$$

$$\sec \theta = -2 \implies \cos \theta = -\frac{1}{2} \implies \theta = 120^\circ, 240^\circ \text{ (QII, QIII, ref } 60^\circ) \quad \sec \theta = 1 \implies \cos \theta = 1 \implies \theta = 0^\circ$$

$$\theta = 0^\circ, 120^\circ, 240^\circ$$

Solve the equation algebraically on the given interval $[0^\circ, 360^\circ)$. (4 points)

14. $2 \cos^2 \theta + \sin \theta = 1$

Two different functions, so use the Pythagorean identity $\cos^2 \theta = 1 - \sin^2 \theta$ to get everything in $\sin \theta$.

$$2(1 - \sin^2 \theta) + \sin \theta - 1 = 0 \implies -2 \sin^2 \theta + \sin \theta + 1 = 0 \implies 2 \sin^2 \theta - \sin \theta - 1 = 0$$

$$(2 \sin \theta + 1)(\sin \theta - 1) = 0 \quad \sin \theta = -\frac{1}{2} \implies \theta = 210^\circ, 330^\circ \quad \sin \theta = 1 \implies \theta = 90^\circ$$

$$\theta = 90^\circ, 210^\circ, 330^\circ$$

Find the exact value of each expression. (1 point each for 15 & 18 and 2 pts each for 16 & 17)

15. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ Range of $\sin^{-1}$ is $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (QI, QIV). Negative input goes to QIV. $\sin \theta = -\frac{\sqrt{3}}{2}, \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \implies \boxed{-\frac{\pi}{3}}$	16. $\cos^{-1}\left(\cos \frac{7\pi}{6}\right)$ $\frac{7\pi}{6}$ is not in $[0, \pi]$, so it cannot be the answer. Work inside out: $\cos \frac{7\pi}{6} = -\frac{\sqrt{3}}{2} \implies \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ Range $[0, \pi]$, negative cosine means QII, reference angle $\frac{\pi}{6}$: $\boxed{\frac{5\pi}{6}}$
17. $\sec^{-1}\left(\csc \frac{\pi}{6}\right)$ Inside first: $\csc \frac{\pi}{6} = \frac{1}{\sin \frac{\pi}{6}} = \frac{1}{1/2} = 2$. $\sec^{-1}(2) = \theta \implies \sec \theta = 2 \implies \cos \theta = \frac{1}{2} \implies \boxed{\frac{\pi}{3}}$ (range of $\sec^{-1}$ is $[0, \pi]$ minus $\frac{\pi}{2}$; a positive input lands in QI)	18. $\cot^{-1}(-\sqrt{3})$ Range of $\cot^{-1}$ is $(0, \pi)$ (QI, QII), so a negative input goes to QII, not QIV. $\cot \theta = -\sqrt{3} \implies \tan \theta = -\frac{1}{\sqrt{3}}, \text{ ref } \frac{\pi}{6}$ $\theta = \pi - \frac{\pi}{6} = \boxed{\frac{5\pi}{6}}$

Give the exact value of each expression. Rationalize the denominator of your answer. (3 points each)

19. $\csc\left(2\sin^{-1}\frac{1}{4}\right)$ Let $u = \sin^{-1}\frac{1}{4}$, so $\sin u = \frac{1}{4}$ (QI). Reference triangle: $y = 1, r = 4, x = \sqrt{16-1} = \sqrt{15}$, so $\cos u = \frac{\sqrt{15}}{4}$. $\csc 2u = \frac{1}{\sin 2u} = \frac{1}{2\sin u \cos u}$ $= \frac{1}{2\left(\frac{1}{4}\right)\left(\frac{\sqrt{15}}{4}\right)} = \frac{1}{\frac{\sqrt{15}}{8}} = \frac{8}{\sqrt{15}}$ Rationalize: $\frac{8}{\sqrt{15}} \cdot \frac{\sqrt{15}}{\sqrt{15}} = \boxed{\frac{8\sqrt{15}}{15}}$	20. $\sin\left(\tan^{-1}2 + \cos^{-1}\frac{3}{5}\right)$ $\sin(A+B) = \sin A \cos B + \cos A \sin B$. $A = \tan^{-1}2$: $\tan A = \frac{2}{1}$, QI, $r = \sqrt{5}$, so $\sin A = \frac{2}{\sqrt{5}}$, $\cos A = \frac{1}{\sqrt{5}}$. $B = \cos^{-1}\frac{3}{5}$: $x = 3, r = 5, y = 4$, so $\cos B = \frac{3}{5}$, $\sin B = \frac{4}{5}$. $\left(\frac{2}{\sqrt{5}}\right)\left(\frac{3}{5}\right) + \left(\frac{1}{\sqrt{5}}\right)\left(\frac{4}{5}\right)$ $= \frac{6}{5\sqrt{5}} + \frac{4}{5\sqrt{5}} = \frac{10}{5\sqrt{5}} = \frac{2}{\sqrt{5}}$ Rationalize: $\frac{2}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \boxed{\frac{2\sqrt{5}}{5}}$
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